

## Mid-term Exam (Graph Neural Networks - Fall 2026)

Full Name:

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Note: Students should write in English for evaluation.

**9. (Node2vec)** Consider an undirected graph  $G$  given in the following figure. A random walk has just moved from node  $A$  to node  $B$ , with  $p = 2$  and  $q = 0.5$ .

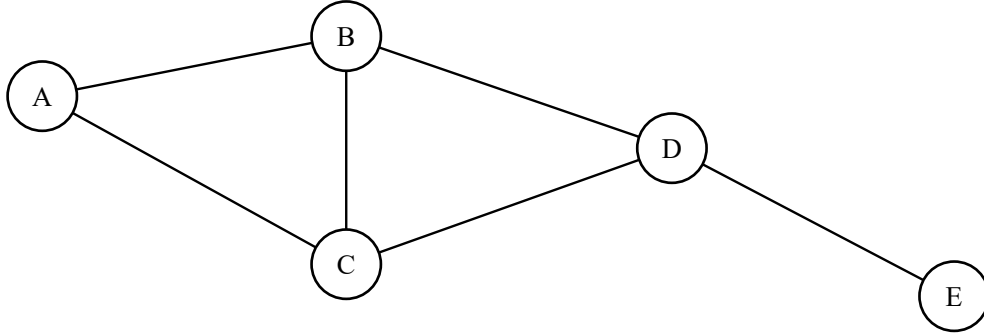


Figure 1. Graph G

The unnormalized transition probability from  $B$  to a next node  $x$  is  $\pi_{Bx} = \alpha_{pq}(A, x)$ , where

$$\alpha_{pq}(A, x) = 1/p \text{ if } d(A, x) = 0; \quad 1 \text{ if } d(A, x) = 1; \quad 1/q \text{ if } d(A, x) = 2$$

and  $d(A, x)$  is the shortest path distance between node  $A$  and node  $x$ .

- a) Explain BFS and DFS, and explain which one a small  $q$  makes the walk behave like.
- b) List all possible next nodes from node  $B$  and calculate the unnormalized transition probabilities  $\pi_{Bx}$ .
- c) Calculate the normalized transition probabilities.

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**10. (LINE)** Consider an undirected graph  $G$  with node embeddings

$$u_A = 1, u_B = 0.5, u_C = -0.5, u_D = -1$$

and context embeddings

$$u'_A = 0.5, u'_B = 1, u'_C = -1, u'_D = -0.5.$$

The second-order proximity is defined as

$$p_2(j|x) = \exp(u'_j u_x) / \sum_k \exp(u'_k u_x).$$

Given  $e^{0.25} = 1.28$ ,  $e^{0.5} = 1.65$ ,  $e^{-0.25} = 0.78$ ,  $e^{-0.5} = 0.61$ .

- a) What is second-order proximity, and how does it differ from first-order proximity?
- b) Calculate  $p_2(A|B)$ ,  $p_2(B|B)$ ,  $p_2(C|B)$  and  $p_2(D|B)$ .

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**11. (GCN)** Given a graph with an adjacency matrix  $A$  and initial node feature matrix  $H^{(0)}$  as follows:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 0 \\ 3 \end{bmatrix}$$

Assume that the hidden layer of a GCN model of all nodes at layer  $(k)$  can be calculated as:

$$H^{(k)} = \sigma(\tilde{A} \cdot H^{(k-1)}),$$

where  $H^{(k)}$  denotes the output at layer  $k$ ,  $\tilde{A}$  is the normalized matrix ( $\tilde{A} = D^{-1}A$ ), and  $\sigma$  is a ReLU function  $\text{ReLU}(x) = \max(0, x)$ .

- a) Explain how information is propagated in a GCN layer.
- b) Calculate  $\tilde{A}$ .
- c) Calculate the output of the GCN model at layer  $k = 1$ .

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**12. (GraphSAGE)** Consider an undirected graph  $G$  of five nodes A, B, C, D and E given in the following figure. Each node has initial features that are the numbers standing next to it (i.e. the initial feature of node 'A' is  $h_A^{(0)} = 0.4$ ). According to the GraphSAGE model with an AGGREGATE that is a MEAN function, the feature of a node  $i$  at layer  $k$  can be updated as:

$$h_{N(i)}^{(k)} = \text{AGGREGATE}(\{ h_u^{(k-1)}, \forall u \in N(i) \})$$

$$h_i^{(k)} = \text{ReLU}( h_i^{(k-1)} \parallel h_{N(i)}^{(k)} )$$

where  $\parallel$  is a concatenation,  $\text{ReLU}(x) = \max(0, x)$ , and  $N(i)$  is the set of neighbour nodes of node  $i$ .

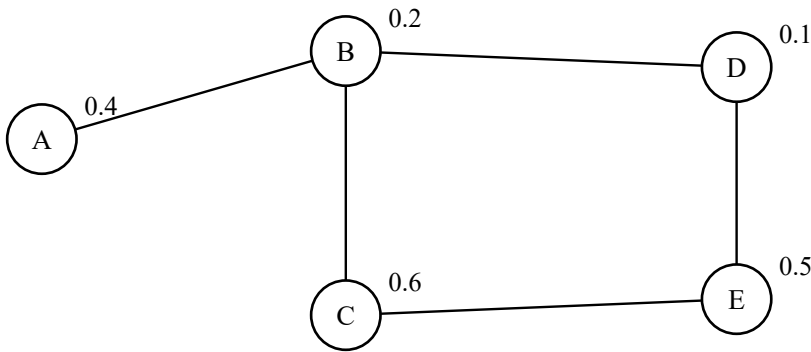


Figure 2. Graph G

- What is the main idea behind the GraphSAGE model, and how does it differ from traditional GCNs?
- Calculate the feature of each node at  $k = 1$ .
- Calculate a graph-level embedding  $h_G$  by using a 'Mean' global pooling when  $k = 1$ .

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**13. (ClusterGCN)** Consider an undirected graph  $G$  of six nodes A, B, C, D, E and F given in the following figure. The graph  $G$  contains two clusters  $G_1$  and  $G_2$ . Each node has initial features that are the numbers standing next to it. According to the ClusterGCN model, the feature of a node  $i$  at layer  $k$  can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \forall u \in N(i), G_u = G_i \})$$

$$h_i^{(k)} = \text{ReLU}( h_i^{(k-1)} \parallel h_{N(i)}^{(k)} )$$

where  $\parallel$  is a concatenation and  $\text{ReLU}(x) = \max(0, x)$ .

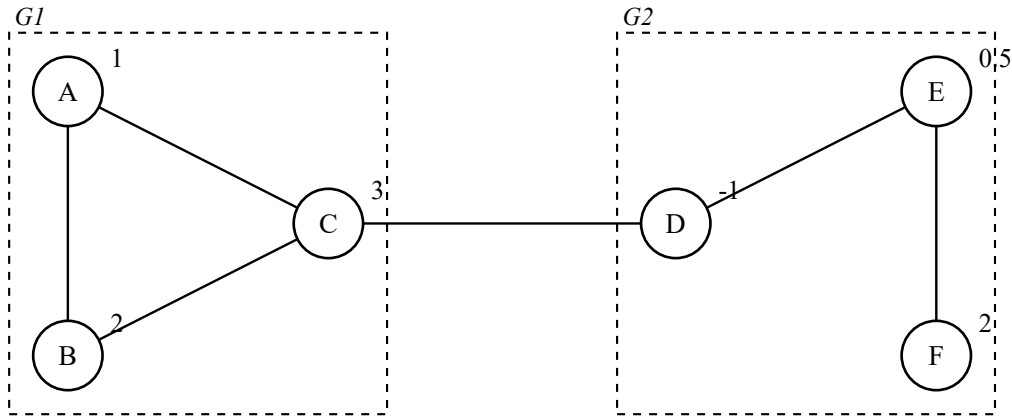


Figure 3. Graph G with two clusters

- What is mini-batch training, and why does ClusterGCN need it?
- Calculate the output representations of all nodes at layer  $k = 1$ .

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**14. (GraphSAINT)** Consider an undirected graph  $G$  of six nodes A, B, C, D, E and F given in the following figure. The graph  $G$  has a sampled subgraph  $G_1$ . Each node has initial features that are the numbers standing next to it. According to the GraphSAINT model, the feature of a node  $i$  at layer  $k$  can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \forall u \in N(i), G_u = G_i \})$$

$$h_i^{(k)} = \text{ReLU}( h_i^{(k-1)} \parallel h_{N(i)}^{(k)} )$$

where  $\parallel$  is a concatenation and  $\text{ReLU}(x) = \max(0, x)$ .

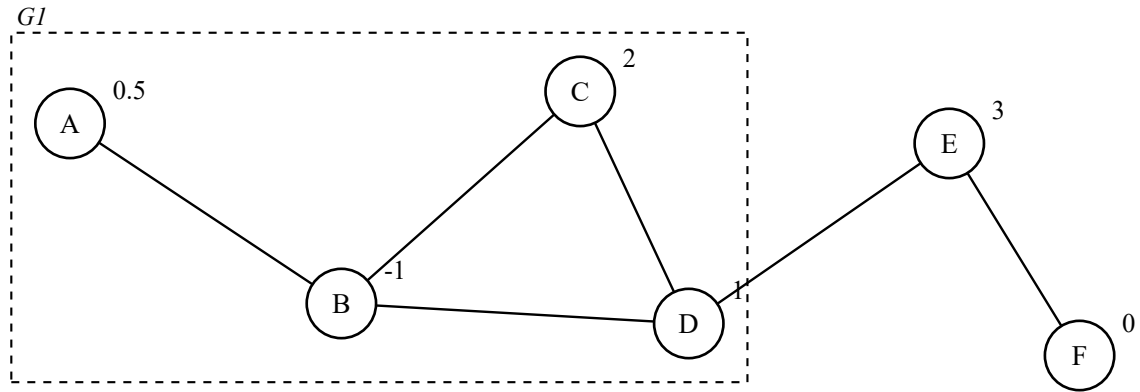


Figure 4. Graph  $G$  with a sampled subgraph  $G_1$

- a) How does GraphSAINT's sampling strategy differ from node-wise or layer-wise sampling methods like GraphSAGE and FastGCN?
- b) Calculate the output representations of nodes A, B, C and D at layer  $k = 1$ .

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**15. (JK Network)** Given a graph with an adjacency matrix  $A$  and initial node feature matrix  $H^{(0)}$  as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} -1 \\ 0 \\ 1 \\ -1 \\ 2 \end{bmatrix}$$

Assume that the output of a JK network model of all nodes at layer  $(k)$  can be calculated as:

$$H^{(k)} = \max( \sigma(\tilde{A} \cdot H^{(0)}), \sigma(\tilde{A} \cdot H^{(1)}), \dots, \sigma(\tilde{A} \cdot H^{(k-1)}) )$$

where  $H^{(k)}$  denotes the output at layer  $k$ ,  $\tilde{A}$  is the normalized matrix ( $\tilde{A} = D^{-1}A$ ), and  $\sigma$  is a ReLU function  $\text{ReLU}(x) = \max(0, x)$ .

- a) Explain how Jumping Knowledge Networks help mitigate the over-smoothing problem in deep GCNs.
- b) Calculate  $\tilde{A}$ .
- c) Calculate the output representations at layer  $k = 2$ .

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**16. (GCNII)** Given a graph with an adjacency matrix  $A$  and initial node feature matrix  $H^{(0)}$  as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 3 \\ -1 \\ 1 \\ 2 \\ -3 \end{bmatrix}$$

Assume that the output of a GCNII model of all nodes at layer  $(k)$  can be calculated as:

$$H^{(k)} = \sigma[ ((1 - \beta) I_n) \cdot ((1 - \alpha) \tilde{A} \cdot H^{(k-1)} + \alpha H^{(0)}) ]$$

where  $H^{(k)}$  denotes the output at layer  $k$ ,  $\tilde{A}$  is the normalized matrix ( $\tilde{A} = D^{-1}A$ ),  $I_n$  is the identity matrix,  $\alpha = \beta = 0.5$ , and  $\sigma$  is a ReLU function  $\text{ReLU}(x) = \max(0, x)$ .

- a) What are the two major components introduced in GCNII to overcome the depth limitation problem in GCNs?
- b) Calculate  $\tilde{A}$ .
- c) Calculate the output representations at layer  $k = 1$ .



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**17. (DeepGCN)** Given a graph with an adjacency matrix  $A$  and initial node feature matrix  $H^{(0)}$  as follows:

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 2 \\ -2 \\ 1 \\ 0 \end{bmatrix}$$

Assume that the output of a DeepGCN model with a residual connection of all nodes at layer  $(k)$  can be calculated as:

$$H^{(k)} = \sigma(\tilde{A} \cdot H^{(k-1)}) + H^{(k-1)}$$

where  $\tilde{A} = D^{-1}A$  and  $\sigma$  is a ReLU function  $\text{ReLU}(x) = \max(0, x)$ .

- a) Explain why a residual connection allows a GCN to be made deeper.
- b) Calculate the output representations at layer  $k = 1$  and  $k = 2$ .
- c) Compare  $H^{(1)}$  with the output of a plain GCN without the residual connection.

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**1. (Relational Classification)** Consider an undirected graph  $G$  of five nodes given in the following figure. Nodes 1 and 5 are labelled ( $P(Y_1) = 1, P(Y_5) = 0$ ) and the others are unlabelled and initialized to 0.5. The relational classifier updates every unlabelled node as:

$$P(Y_v = c) = ( \sum_{(v,u) \in E} A_{v,u} P(Y_u = c) ) / ( \sum_{(v,u) \in E} A_{v,u} )$$

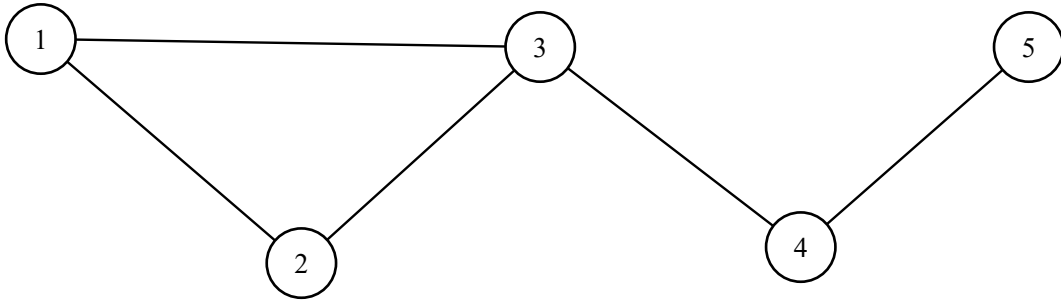


Figure 5. Graph G

- a) Explain the key assumption behind relational classification, and state one limitation of it.
- b) Calculate  $P(Y_v)$  for every unlabelled node after the 1st iteration.
- c) Calculate  $P(Y_v)$  after the 2nd iteration. Update nodes in the order 2, 3, 4 using the values from the previous iteration only.

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**2. (FastGCN)** Consider the undirected graph  $G$  with adjacency matrix  $A$  and node feature vector  $h^{(0)}$  below. FastGCN replaces the full neighbourhood sum of a GCN layer by a layer-wise Monte-Carlo estimate:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$h^{(0)} = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \\ 3 \end{bmatrix}$$

$$h^{(l+1)}(v) = \sigma \left( \frac{1}{t_l} \sum_{j=1}^{t_l} \tilde{A}(v, u_j) h^{(l)}(u_j) \right)$$

where  $\tilde{A} = D^{-1}A$ , the  $t_l$  nodes  $u_j$  are drawn once per layer from the importance distribution  $q(u) \propto \|\tilde{A}(:, u)\|^2$ , and  $\text{ReLU}(x) = \max(0, x)$ .

- a) How does layer-wise sampling in FastGCN differ from the node-wise sampling of GraphSAGE, and what problem does it solve?
- b) Calculate the importance distribution  $q(u)$  over the five nodes.
- c) Suppose the sampled set for the layer is  $\{B, C\}$  ( $t_l = 2$ ). Calculate the estimate of  $h^{(1)}(A)$ .

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**3. (ASGCN)** Consider the same graph  $G$  as in question 11, with  $\tilde{A} = D^{-1}A$  and the node feature vector  $h^{(0)}$  below. ASGCN rewrites a GCN layer as an expectation:

$$h^{(0)} = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \\ 3 \end{bmatrix}$$

$$h^{(l+1)}(v_i) = \sigma( W^{(l)} N(v_i) E_{p(u_j | v_i)}[ h^{(l)}(u_j) ] )$$

where  $N(v_i) = \sum_j \tilde{A}_{i,j}$  is the normalization factor and  $p(u_j | v_i) = \tilde{A}_{i,j} / N(v_i)$  is the probability of sampling node  $u_j$  given the parent node  $v_i$ . Take  $W^{(l)} = 1$  and  $\text{ReLU}(x) = \max(0, x)$ .

- a) ASGCN is called 'adaptive' sampling. What does the proposal distribution depend on in ASGCN that it does not depend on in FastGCN?
- b) Calculate  $N(v_i)$  and  $p(u_j | v_i)$  for the parent node  $v_i = B$ .
- c) Calculate  $E_{p(u_j | B)}[h^{(0)}(u_j)]$  and the resulting  $h^{(1)}(B)$ .

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**4. (APPNP)** Given a graph with an adjacency matrix  $A$  and a prediction matrix  $H$  produced by an MLP as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

APPNP (Approximate Personalized Propagation of Neural Predictions) propagates the predictions with a teleport probability  $\alpha$ :

$$Z^{(0)} = H, \quad Z^{(k+1)} = (1 - \alpha) \tilde{A} Z^{(k)} + \alpha H$$

where  $\tilde{A} = D^{-1}A$  and  $\alpha = 0.5$ . No nonlinearity is applied between propagation steps.

- a) Explain the teleport mechanism of Personalized PageRank and why it prevents over-smoothing even with many propagation steps.
- b) Calculate  $\tilde{A}$  and  $Z^{(1)}$ .
- c) Calculate  $Z^{(2)}$ .

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**5. (GPR-GNN)** Given a graph with an adjacency matrix  $A$  and hidden features  $H^{(0)}$  produced by a neural network as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 2 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

GPR-GNN (Generalized PageRank GNN) separates feature extraction from propagation:

$$H^{(k)} = \tilde{A} H^{(k-1)}, \quad Z = \sum_{k=0}^K \gamma_k H^{(k)}$$

where  $\tilde{A} = D^{-1}A$ , the weights  $\gamma_k$  are learned, and here  $K = 2$  with  $\gamma_0 = 0.5$ ,  $\gamma_1 = 0.3$ ,  $\gamma_2 = -0.2$ .

- a) What can GPR-GNN do that APPNP cannot, and why does allowing negative  $\gamma_k$  matter for heterophilic graphs?
- b) Calculate  $H^{(1)}$  and  $H^{(2)}$ .
- c) Calculate  $Z$ .

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**6. (MixHop)** Given a graph with an adjacency matrix  $A$  and initial node feature matrix  $H^{(0)}$  as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}$$

A MixHop layer concatenates several powers of the adjacency matrix:

$$H^{(i+1)} = \mathbb{I}_{j \in P} \sigma(\hat{A}^j H^{(i)} W_j^{(i)})$$

where  $\mathbb{I}$  is a column-wise concatenation,  $P = \{0, 1, 2\}$ ,  $\hat{A} = D^{-1}A$ ,  $W_j^{(i)} = 1$  for every  $j$ , and  $\text{ReLU}(x) = \max(0, x)$ .

- a) State Theorem 1 of MixHop about the two-hop Delta Operator, and explain why a vanilla GCN cannot represent it even with infinitely many layers.
- b) Calculate  $\hat{A}^0 H^{(0)}$ ,  $\hat{A}^1 H^{(0)}$  and  $\hat{A}^2 H^{(0)}$ .
- c) Calculate  $H^{(1)}$ , and calculate the Delta Operator  $\sigma(\hat{A}^0 H^{(0)}) - \sigma(\hat{A}^2 H^{(0)})$ .

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**7. (PairNorm)** After one GCN layer the node features of a four-node graph are

$$X = \begin{bmatrix} 3 & 1 \\ 1 & -1 \\ -1 & 2 \\ 1 & 2 \end{bmatrix}$$

PairNorm normalizes them in two steps, a centering step and a scaling step:

$$x_i^c = x_i - (1/n) \sum_{i=1}^n x_i \quad (\text{Center})$$

$$x_i = s \cdot x_i^c / \sqrt{(1/n) \sum_{i=1}^n \|x_i^c\|_2^2} \quad (\text{Scale})$$

with  $s = 1$  and  $n = 4$ .

- a) Explain node-wise and feature-wise over-smoothing, and explain which quantity PairNorm keeps constant across layers.
- b) Calculate the centered features  $x_i^c$ .
- c) Calculate the scaling denominator and the final  $x_i$ .



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**8. (DropEdge)** Consider the complete graph  $K_4$  on nodes A, B, C and D with the adjacency matrix  $A$  and initial node feature matrix  $H^{(0)}$  below.

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 2 \\ 4 \\ -2 \\ 0 \end{bmatrix}$$

DropEdge keeps each undirected edge independently with probability  $1 - p$  at every training epoch, and the layer is then computed on the surviving graph:

$$H^{(1)} = \sigma( (D^{(t)})^{-1} A^{(t)} H^{(0)} )$$

where  $A^{(t)}$  and  $D^{(t)}$  are rebuilt from the surviving edges only,  $p = 1/3$ , and  $\text{ReLU}(x) = \max(0, x)$ .

- a) Which two problems does DropEdge target, and how does it differ from Dropout?
- b) Calculate the expected number of surviving edges, and calculate  $H^{(1)}$  on the full graph.
- c) Suppose the edges (A, D) and (B, C) are dropped in this epoch. Recalculate  $H^{(1)}$ .